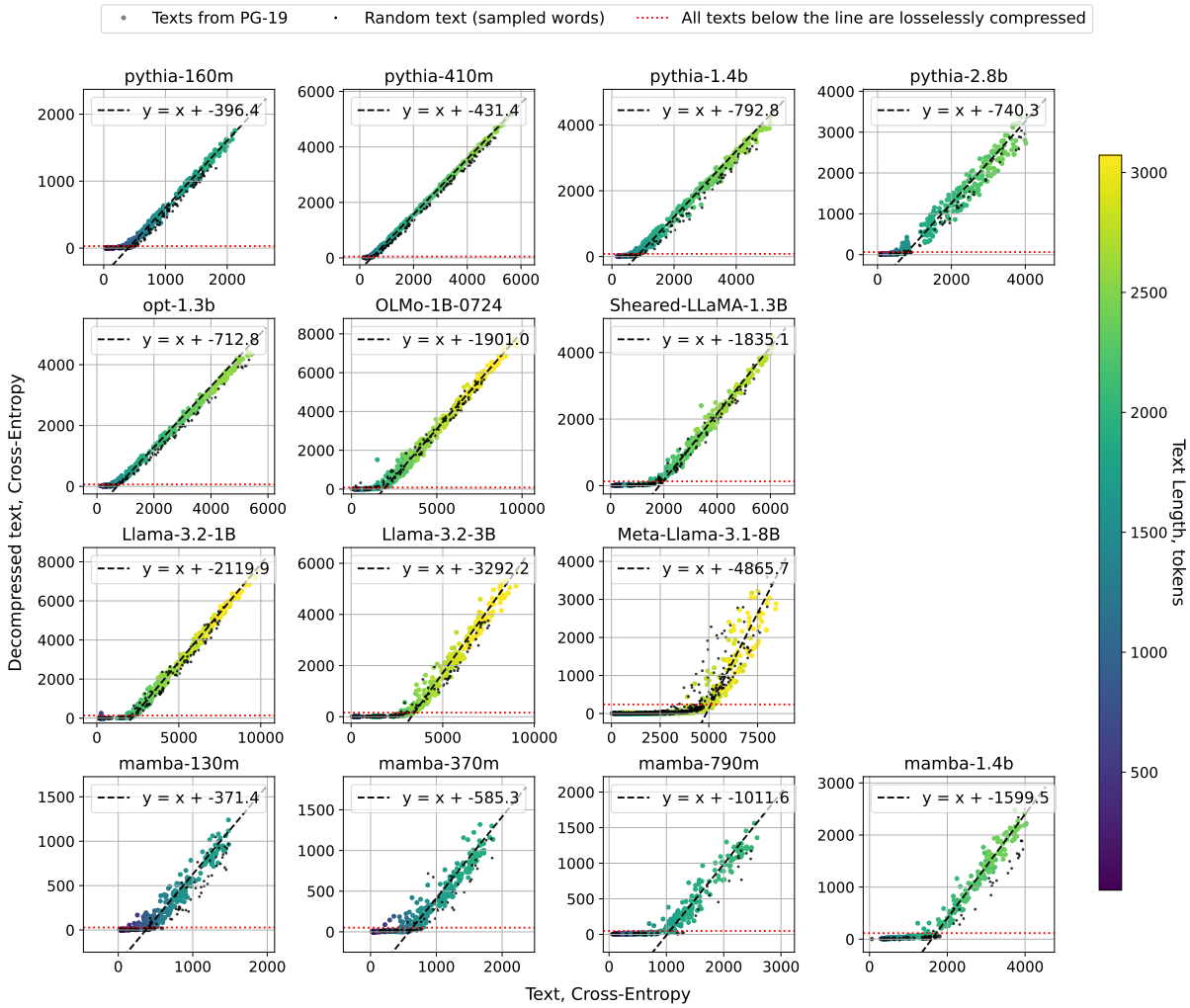

Figure 6: **Information gain of text compression to `[mem]` vector doesn't depend on language understanding capabilities of models.** Compression results for various language models show the relationship between the cross-entropy (CE) of the original and decompressed texts. If the text CE falls below a model-specific threshold (red line), the text is losslessly compressed. This value is a input vector capacity in terms of entropy (*Information Gain*, C_H). For texts that are not perfectly compressed, the compression process reduces their CE to a consistent, model-specific value (bias of the black dashed line). Larger models (e.g., Llama-3.1-8B) can handle longer texts before reaching the compression threshold, due to their greater capacity compared to smaller models (e.g., Pythia-160M). This behavior holds for both natural texts (*PG-19*) and unnatural *random* texts consisting of random word sequences. The result is not limited to transformer-based architectures: non-transformer models, such as *Mamba*, exhibit the same pattern.

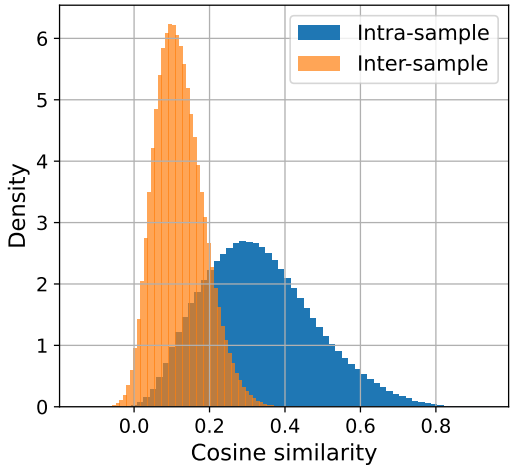

Figure 7: **Intra/inter-sample embeddings cosine similarity.** Empirical probability densities of cosine similarity between intra-sample and inter-sample embeddings. Intra-sample similarities are measured between of the same sequence of tokens, while inter-sample between different ones. Measured on GovReport (Huang et al., 2021) and Sheared-Llama-1.3B (Xia et al., 2024).

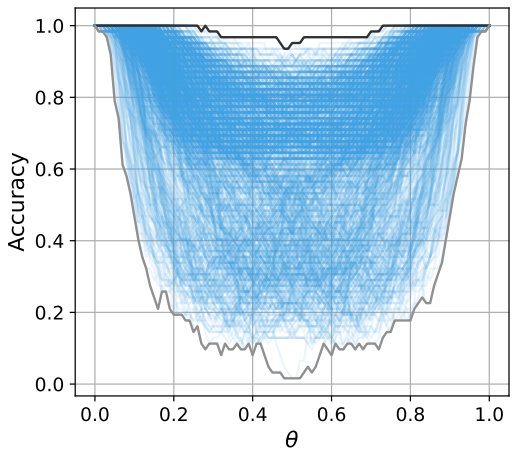

Figure 8: **Intra-sample Interpolation Accuracies.** Interpolation lines are provided for all pairs between 32 embeddings of the same input sequence. All interpolation lines are printed with high transparency to show denser regions. Grey lines depict minimums and maximums of the accuracy for a given interpolation parameter θ.